

AIKEN, SC, USA

WMO: 720601

Lat: 33.650N

Lon: 81.683W

Elev: 161

StdP: 99.40

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00193

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-2.6	-13.7	1.2	-2.6	-10.9	1.5	1.6	9.5	10.5	8.8	11.8	1.5	40	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	35.2	24.6	33.9	24.3	32.7	24.2	28.1	32.4	27.2	30.4	26.0	29.3	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.5	30.5	26.2	22.0	29.2	25.1	20.7	27.6	91.2	32.1	86.4	30.0	81.8	28.6	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.2	5.9	DB	-8.1	37.0	2.0	1.7	-9.5	38.2	-10.7	39.2	-11.9	40.1	-13.3	41.3
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	6.6	9.4	12.4	17.3	21.9	25.8	27.1	26.2	23.8	17.8	11.3	8.9
	DBStd	8.20	5.51	5.19	5.06	4.01	3.29	2.27	1.81	1.88	2.77	4.11	4.12	4.99
	HDD10.0	359	138	70	34	2	0	0	0	0	0	1	32	83
	HDD18.3	1455	366	253	192	66	11	0	0	0	2	59	213	294
	CDD10.0	3063	31	53	109	222	370	474	529	502	413	242	71	49
	CDD18.3	1119	1	2	9	35	122	224	271	243	165	42	2	2
	CDH23.3	10416	4	14	83	378	1092	2201	2669	2194	1395	363	18	3
	CDH26.7	4411	0	0	8	81	395	1032	1280	967	564	83	1	0
Wind	WSAvg	2.5	2.7	2.9	3.0	3.0	2.6	2.3	2.4	2.0	2.2	2.3	2.3	2.5
Precipitation	PrecAvg	1326	130	115	138	93	101	138	136	134	101	75	75	93
	PrecMax	1962	263	224	367	340	204	350	298	291	216	323	292	192
	PrecMin	896	27	29	27	19	9	12	35	21	2	0	4	21
	PrecStd	223	61	51	73	65	50	66	66	69	61	66	54	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	25.9	27.7	31.0	34.8	36.4	37.2	36.0	34.8	31.3	26.0	22.9
		MCWB	18.1	18.4	18.1	21.4	22.3	24.7	27.0	25.5	23.2	22.6	19.2	19.3
	2%	DB	20.1	22.5	26.0	28.8	32.3	35.0	35.3	33.9	32.8	28.7	22.7	21.3
		MCWB	16.1	16.9	17.4	20.7	22.5	24.6	25.6	24.0	23.1	21.6	17.1	18.3
	5%	DB	17.7	20.9	23.0	27.4	31.0	32.9	33.8	32.6	32.1	27.2	21.2	19.0
		MCWB	14.6	15.2	15.9	19.2	21.8	24.1	25.0	24.5	22.6	20.2	16.4	17.0
	10%	DB	16.0	17.7	21.3	25.9	28.8	32.2	32.5	32.0	30.2	25.2	19.2	17.3
		MCWB	12.9	14.0	15.1	18.0	21.3	23.8	25.0	24.4	22.4	19.1	15.7	15.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.5	20.1	24.1	26.2	28.4	29.1	29.0	26.2	25.1	20.8	21.0
		MCDB	22.2	23.1	24.9	28.0	30.0	34.1	34.9	33.0	29.3	26.6	21.6	22.2
	2%	WB	17.6	18.2	19.0	22.3	24.8	27.3	28.3	27.8	25.3	23.4	19.3	19.5
		MCDB	18.7	21.3	23.9	26.4	29.1	32.0	33.1	30.6	29.6	26.3	21.8	20.7
	5%	WB	15.7	17.1	17.8	20.9	23.3	26.1	27.4	26.6	24.5	22.5	17.8	17.5
		MCDB	17.4	19.3	21.2	24.5	27.3	30.1	30.7	28.7	28.7	25.3	19.6	18.0
	10%	WB	13.0	15.4	16.6	19.7	22.4	24.9	26.1	25.5	23.5	21.1	16.5	15.9
		MCDB	14.8	17.5	19.8	23.9	26.5	29.4	29.4	28.8	27.5	23.7	18.5	17.4
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.6	12.7	13.2	11.9	11.5	11.1	10.6	11.3	12.5	12.5	10.2
		MCDBR	13.1	13.6	14.8	14.6	14.2	13.2	12.8	12.1	13.6	13.6	14.2	11.8
		MCWBR	9.0	8.2	7.6	6.8	5.3	4.4	4.2	4.1	4.6	6.5	9.0	8.3
	5% WB	MCDBR	10.5	11.7	12.6	13.0	11.8	12.3	12.0	10.5	11.6	10.8	11.3	8.8
		MCWBR	8.3	7.8	7.1	7.5	5.7	5.6	5.5	5.2	5.1	5.6	8.5	7.1
Clear-Sky Solar Irradiance	taub	0.320	0.327	0.364	0.401	0.465	0.499	0.544	0.525	0.435	0.395	0.348	0.317	
	taud	2.496	2.474	2.398	2.287	2.153	2.102	1.997	2.039	2.276	2.354	2.447	2.538	
	Ebn at Noon	895	925	911	888	831	799	760	768	828	835	851	878	
	Edh at Noon	86	97	113	132	153	160	177	167	127	108	89	78	
All-Sky Solar Radiation	RadAvg	2.80	3.42	4.50	5.67	6.16	6.36	6.07	5.54	4.80	4.02	3.07	2.43	
	RadStd	0.15	0.36	0.33	0.40	0.47	0.45	0.42	0.27	0.41	0.47	0.24	0.23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+184	+129

Nomenclature: See separate page